How the Sovereign Spread Shapes the Snowball Effect: Evidence from Italy

Abstract

This paper investigates the relationship between the Italian sovereign spread (BTP-Bund) and the snowball effect (the interest-growth differential, r-g) over the period 1995–2019. Using OLS regression with HAC standard errors, we examine how the risk premium embedded in the spread translates into changes in debt dynamics, and whether this transmission mechanism differs before and after the euro area sovereign debt crisis. Our baseline results show that higher sovereign spreads are significantly associated with a larger snowball effect. However, the introduction of a post-2010 interaction term reveals a structural break: while spreads had a strong positive effect on r-g before 2010, this relationship weakened substantially in the post-crisis period. We interpret this attenuation as evidence of the stabilizing role played by ECB interventions, particularly following the "whatever it takes" announcement in 2012. A threshold-interaction model further confirms that the sensitivity of the snowball effect to spreads diminished after 2010, with no evidence of convex amplification at high spread levels. These findings suggest that Italy's debt sustainability has depended largely on sovereign risk perceptions and monetary policy conditions, rather than on conventional measures of fiscal discipline alone.

Keywords: Sovereign spreads, snowball effect, debt sustainability, ECB, Italy, structural brea

Introduction and Purposes

The evolution of public debt depends not only on fiscal policy but also on the interaction between growth, interest rates and market confidence (Schroeder, 2015). Central to this dynamic is the snowball effect, the mechanism by which debt increases autonomously when the effective interest rate exceeds economic growth ($r > g$) and declines when the opposite holds ($r < g$) (Blanchard, 2019). This differential represents the mechanical component of debt accumulation, independent of the primary balance, and plays a decisive role in fiscal sustainability (Bohn, 1998; ECB, 2009). In the euro area, several debt reduction episodes, such as in Belgium and Finland during the 1990s, were driven by favorable interest-growth differentials, while countries like Italy, facing persistently higher rates and market volatility, experienced prolonged debt accumulation (ECB, 2009). The Italian structural fiscal fragility and high debt levels makes it a good example to understand how market conditions, indexed by the spread[1] on bonds (main explanatory variable) influence the snowball effect (dependent variable).

The goal is not to investigate whether markets "discipline" fiscal fundamentals – as in the De Grauwe-type literature, where the spread is a function of debt, deficit and growth – but rather to examine the extent to which the risk premium embedded in the spread actually translates into a deterioration of r-g and how this channel changes between the pre-crisis period and the period following the ECB's intervention (Corsetti et al. (2015)). It is important to note that the analysis covers the period 1995–2019 and relies on annual data. The choice of the sample period is motivated by two main considerations. First, consistent and complete data for all variables are available only from 1995 onward. Second, the post-2019 period is excluded because the COVID-19 shock introduces dynamics that are not informative for the mechanism under study.

[1] The sovereign spread is constructed using 10-year Italian and German yields from the ECB Statistical Data Warehouse, while all other variables are sourced from the AMECO database.

Specifically, the research done here investigates whether the Italian BTP[2]–Bund spread significantly affects the snowball component of debt dynamics, alongside fiscal and macroeconomic controls such as the debt ratio, primary balance, growth and inflation (Blanchard, 2019). Differently form (De Grauwe and Ji, 2013), we use a simpler but effective method to model the effects of a pre-post 2010 interaction, which is the introduction of a post-2010 dummy variable.

Beyond the baseline linear specification, the analysis also tests for nonlinear behavior through the usage of an index function, examining whether the effect of the spread becomes disproportionately stronger, or weaker, once it exceeds a critical threshold.

While this analysis focuses on sovereign risk and market transmission mechanisms in Italy, it is important to note that other types of country risk, such as political and institutional risks, can indirectly influence sovereign spreads and the snowball effect. Political uncertainty (unstable coalitions, electoral volatility) may elevate risk premia even without changes in fiscal fundamentals, as documented during the 2018 budget tensions (Balduzzi et al., 2020). Likewise, weak institutional capacity or inconsistent economic policy frameworks can undermine investor confidence, amplifying the impact of sovereign risk on debt dynamics. These additional layers of country risk interact with market perceptions and monetary backstops, potentially exacerbating or dampening the transmission studied here.

Discussion

In this section, we interpret the regression findings on the relationship between Italy's sovereign spreads and its snowball effect (the interest–growth differential), paying special attention to the post-2010 period as the results consistently indicate a structural change around 2010 and confirm intuitive macroeconomic dynamics. To frame the discussion, it is useful to note the magnitude of the variables in question

	Snowball	Spread	Debt_GDP	PrimBal	Growth	Inflation
Mean	2.4407	1.2612	117.49	-2.786	0.5816	1.898
Std	2.1702	1.1115	11.3	1.6062	1.8938	1.094
Min	-0.0007	0.1775	103.46	-6.1396	-5.4511	-0.45634
Max	8.3687	3.9975	134.8	-0.2002	3.8087	3.843
Median	1.7308	1.1546	116.31	-3.0221	1.0552	2.0358

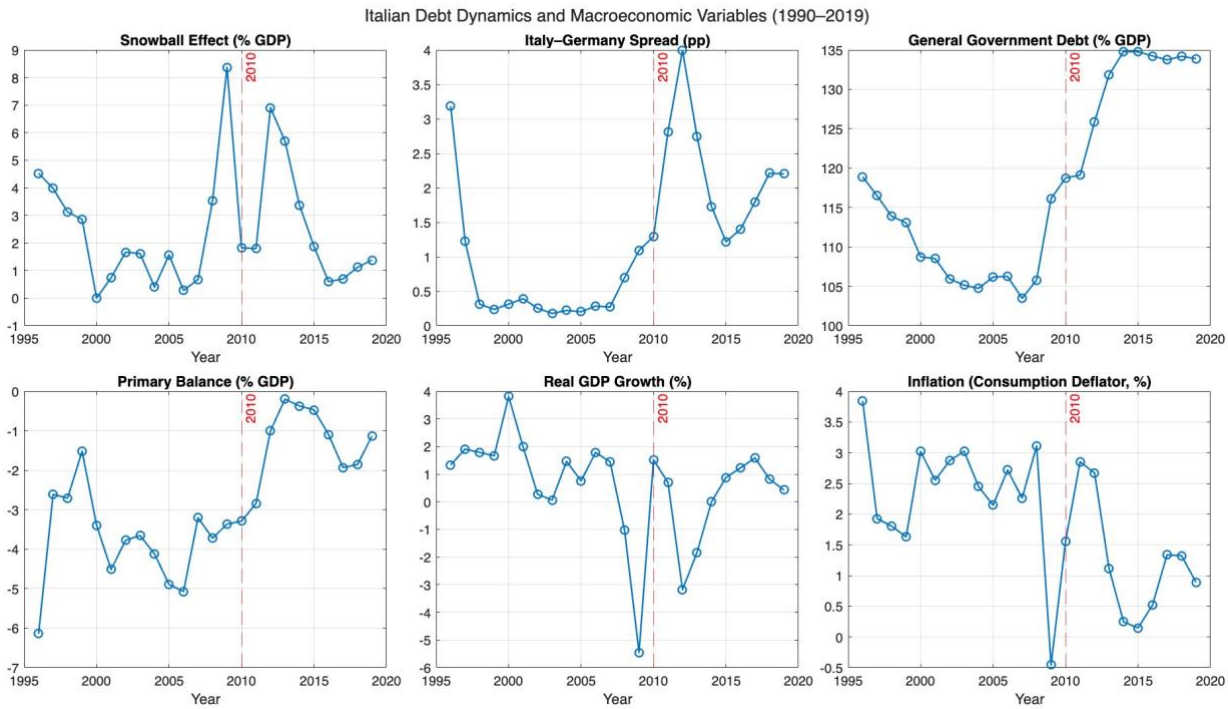
The snowball effect $sb_t = r_t - g_t$ displays a positive mean (2.44) and a sizeable dispersion (sd = 2.17), ranging from near-zero values to peaks above 8.3, confirming that episodes of elevated interest-growth differentials are recurrent and sometimes substantial over the sample.

The sovereign spread, the key explanatory variable, averages 1.26 percentage points with a standard deviation of 1.11. Its range (0.18 to 3.99) reflects pronounced shifts in market perceptions of sovereign risk, particularly around periods of financial stress.

Fiscal and macroeconomic controls exhibit notable variability. Debt-to-GDP remains persistently high (mean 117.5%), while the primary balance is on average moderately negative (–2.79), with considerable cyclical variation. Real GDP growth shows wide fluctuations (mean 0.58%, sd = 1.89), from deep contractions to robust expansions, whereas inflation remains more stable, averaging 1.90%.

The descriptive evidence indicates a sample characterized by substantial macro-financial volatility, consistent with the Italian case. This provides a suitable environment to examine how movements in sovereign spreads translate into changes in the interest-growth differential, and whether this transmission mechanism varies across policy regimes.

[2] Beni del Tesoro Poliennali



Author's calculations based on AMECO macroeconomic and fiscal data and ECB SDW bond yield series for Italy and Germany (10-year maturity)

The models

Baseline model

In the analysis many models were tried based on the robustness checks. The simpler empirical specification that relates the snowball effect to financial, fiscal and macroeconomic conditions in Italy, and is expressed as:

$$sb_t = \alpha + \beta 1 \cdot spr_t + \beta 2 \cdot b_t{-}1 + \beta 3 \cdot pb_t + \beta 4 \cdot g_t + \beta 5 \cdot \pi_t + \varepsilon_t$$

where *sb_t* denotes the snowball effect, *spr_t* the yield spread**,** *b_t-1* he lagged debt ratio, *pb_t* the cyclically adjusted primary balance, *g_t* real GDP growth, and *π_t* inflation.

	Estimate	SE_OLS	t_OLS	p_OLS	SE_HAC	t_HAC	p_HAC
const	16.242	9.0946	1.7859	0.090966	7.5742	2.1444	0.045898
spr	1.1781	0.57136	2.0619	0.053955	0.45064	2.6143	0.017561
debtLag	−0.11481	0.073954	−1.5524	0.13798	0.060062	−1.9114	0.072003
pb	0.13414	0.28645	0.46829	0.6452	0.29031	0.46207	0.64957
g	−0.47338	0.29902	−1.5831	0.13081	0.26149	−1.8103	0.086979
pi	−0.60596	0.54038	−1.1214	0.27687	0.23471	−2.5817	0.018807

In the simpler baseline linear model, higher sovereign spreads are associated with a higher snowball effect. The findings even from here suggest that the relationship we are looking for exists and is statistically significant. The sovereign spread displays a positive and statistically significant coefficient, suggesting that higher financing costs are associated with a larger snowball effect for Italy (Corsetti et al. (2014)). This aligns with well-documented episodes in which increases in market stress, reflected in rising spreads on Italian government securities, have tightened financial conditions and worsened the interest–growth differential (De Grauwe & Ji, 2013; Zoli, 2013). The lagged debt ratio also turns out to be quite significant, although its estimated coefficient is negative, maybe reflecting periods in which Italy's high debt coexisted with favorable interest–growth conditions thereby reducing the real burden of debt despite elevated ratios (Blanchard, 2019). In general, the other fiscal and macroeconomic variables display more mixed patterns, with marginal significance or, in the case of the primary balance, no statistical significance.

The baseline model provides some preliminary evidence of the mechanisms that drive the snowball effect, showing how the financial variables, particularly the sovereign spread and inflation, emerge as the robust determinants, while fiscal variables appear less influential in this simplified setting. At the same time, the limited significance of some coefficients points to potential misspecification or omitted variables, which motivates the exploration of more complex models in the robustness analysis.

Augmented model - SPR×D2010

Consistent with the structural break hypothesis (De Grauwe & Ji, 2013), the empirical analysis reveals robust evidence of a regime shift in the relationship between sovereign spreads and the snowball effect in the Italian case after 2010. This finding supports the view that the euro area sovereign debt crisis, along with the subsequent institutional responses - most notably the ECB's evolving policy stance - fundamentally altered the way markets assessed and priced sovereign risk.
The model is specified as follows:

$$sb_t = \alpha + \beta 1 \cdot spr_t + \beta 2 \cdot spr \times D2010 + \beta 3 \cdot b_t{-}1 + \beta 4 \cdot pb_t + \beta 5 \cdot g_t + \beta 6 \cdot \pi_t + \varepsilon_t$$

where the interaction term spr×D2010 captures the possibility that the marginal effect of spreads on the snowball mechanism changed in the aftermath of the crisis.

	Estimate	SE_OLS	t_OLS	p_OLS	SE_HAC	t_HAC	p_HAC
const	21.649	5.3776	4.0258	0.00087677	3.954	5.4753	4.105e-05
spr	2.6937	0.41808	6.4431	6.0683e-06	0.20078	13.416	1.7935e-10
spr×D2010	-1.9157	0.31942	-5.9976	1.4407e-05	0.18974	-10.097	1.3445e-08
debtLag	-0.14586	0.04342	-3.3593	0.0037216	0.030268	-4.819	0.00016031
pb	0.93681	0.21399	4.3778	0.00041028	0.17946	5.22	6.9331e-05
g	-0.39651	0.17478	-2.2687	0.036601	0.10043	-3.948	0.0010378
pi	-0.48437	0.31565	-1.5345	0.1433	0.12547	-3.8606	0.0012545

The results of the regression indicate that, before 2010, increases in the spread had a strong and statistically significant positive effect on r–g, insofar as they translated into higher borrowing costs relative to economic growth, exacerbating concerns over debt sustainability. However, once the post-2010 period is considered (spr×D2010) the relationship undergoes a notable transformation, as evidenced by the fact that this term turns strongly negative and remains highly significant ($p < 0.01$), which suggests the emergence of a new regime (De Grauwe and Ji (2013)). In fact as the combined effect of the spread and its interaction is taken into account, it becomes clear that the slope of the spread's influence on r–g flattens considerably after 2010, indicating that, under the altered policy and market conditions that prevailed in the post-crisis era, the snowball effect in Italy became significantly less responsive to fluctuations in sovereign spreads.

This attenuation in transmission aligns with major changes in euro area policy and investor behavior as the period from 2010 onward was marked by a fundamental shift in market sentiment and the adoption of extraordinary monetary policy measures (Pietrunti & Zaghini, 2020). As largely pointed in the economic literature before the GFC[3], financial markets systematically underpriced country-specific risks within the euro area. Spreads across sovereign bonds were compressed and, as documented by Zoli (2013), from 1999 to 2007 they were largely driven by common global risk factors rather than by differences in fiscal fundamentals. In that environment, even highly indebted and institutionally fragile issuers such as Italy, Spain or Portugal traded at yields only marginally above German Bunds, despite markedly weaker debt positions as investors behaved as if membership in the monetary union had effectively pooled sovereign risk (Zoli, 2013).

With the global financial crisis and the subsequent euro-area sovereign debt turmoil this complacency unravelled. As banking and macroeconomic stress migrated to public balance sheets investors became far more

[3] Great Financial Crisis

discerning and began demanding higher risk premia from fiscally vulnerable states. A central framework in this regard is the sovereign risk channel developed by Corsetti et al. (2014). According to this mechanism, an increase in sovereign risk premia is not merely a reflection of deteriorating fundamentals but can itself become a source of macroeconomic fragility.

Our regression results, however, show a different pattern: while spreads remain positively related to the snowball effect, the interaction term spr×D2010 enters with a large and significantly negative coefficient. In other words, the marginal impact of an increase in spreads on sb is robustly positive t before 2010 but becomes much weaker in the post-2010 period. This could seem counterintuitive but becomes clearer in light of the forceful and credibility-enhancing responses by the ECB (Corsetti et al. (2015), Zettelmeyer et al. (2013)).
The crucial turning point came in 2012, when the Italian spread over the German bund reached its peak and the ECB President Mario Draghi pledged to do "whatever it takes" to preserve the euro. This announcement, followed by the announcement of OMT[4] program and other subsequently[5], marked a turning point in the ECB's institutional role, effectively positioning it as a lender of last resort for sovereigns and sharply reducing perceived tail risks in euro area bond market. Empirical work by De Grauwe and Ji (2014) and Pietrunti and Zaghini (2020) shows that, after these interventions, the sensitivity of peripheral spreads to self-fulfilling market sentiment declined, and the link between spreads and liquidity or redenomination fears was significantly attenuated.

The estimates found in the regression fit neatly into this narrative. A positive and significant coefficient on *spr_t* captures the pre-crisis and early-crisis regime, where higher spreads translated almost one-for-one into a higher snowball effect and, therefore, into stronger debt-dynamic pressures. The strongly negative coefficient on spr×D2010 indicates that, after 2010, this pass-through was substantially reduced: spreads still reflected perceived sovereign risk, but their ability to mechanically raise the interest–growth differential was muted, becoming structurally weaker. While increases in spreads continued to reflect perceived sovereign risk, their capacity to mechanically transmit into higher r-g, and thereby intensify debt sustainability pressures, was markedly reduced in the post-crisis period. This shift suggests that, although market discrimination among issuers persisted or even intensified, the effective impact of spreads on public financing costs changed as market discipline and risk discrimination started operating within an institutional framework in which the ECB's backstop and an environment of very low policy rates dampened the impact of market tensions on actual borrowing costs.

The broader implication is that after 2012, Italy became somewhat shielded from self-fulfilling debt spirals: even if spreads rose due to market concerns, their impact on borrowing costs was muted by the ECB's presence. This breaking of the so-called "doom loop", where default fears feed higher spreads, which in turn raise borrowing costs and thus the risk of default, was a central policy achievement of the euro crisis response (De Grauwe and Ji, 2014; Zoli, 2013).

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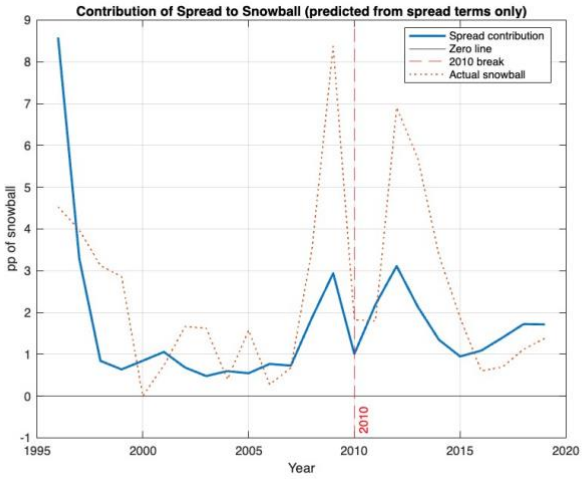
[4] Outright Monetary Transactions

[5] CBPP1, CBPP2, SMP, CBPP3, ABSPP, APP represent some successive ECB interventions targeting covered bonds, sovereign bonds and asset-backed securities



Author's calculations based on AMECO macroeconomic and fiscal data and ECB SDW bond yield series for Italy and Germany (10-year maturity)

The extended regression results confirm that domestic fundamentals significantly influenced Italy's interest-growth differential over 1995–2019. Contrary to standard theoretical expectations, the coefficient on lagged public debt is negative and highly significant, suggesting that higher debt levels were historically associated with a reduction in the snowball effect. Although counterintuitive, this result may reflect Italy's specific context, marked by external monetary interventions and a setting in which high debt did not automatically translate into upward pressure on interest rates. This interpretation is supported by studies such as Bacchiocchi & Borghi (2016), which emphasize the central role of European monetary policy in containing sovereign risk premia for Italy. Similarly, the positive coefficient on the primary balance diverges from theoretical expectations: a higher primary surplus is associated with an increase in the snowball effect. This outcome may be driven by a procyclical dynamic in which fiscal adjustment may occur during episodes of financial stress, when nominal growth is weak and real interest rates are elevated, thereby offsetting the benefits for debt dynamics. Consistently, the highly significant and robust coefficients on the spread and its interaction with the post-2010 dummy confirm that in the Italian context, debt sustainability has depended largely on sovereign risk perceptions and monetary policy conditions, rather than on conventional measures of fiscal discipline. Finally, the coefficients associated with real GDP growth and inflation are both negative and statistically significant, confirming the theoretical intuition that stronger nominal growth contributes to reducing the r–g differential through both mechanical effects and lower credit risk. This is in line with the findings of Blanchard (2019), who argues that sufficiently high growth rates can help keep debt dynamics under control.

Taken together, these results suggest that Italy's experience diverges from the standard sustainability narrative based solely on fiscal discipline and highlights instead, again, the critical importance of European monetary and financial conditions, market expectations, and ECB interventions in shaping the actual trajectory of public debt.

A non-linear trial

We extend the analysis of the spread–snowball effect relationship by estimating a threshold-interaction model that captures both potential nonlinearities and structural breaks around 2010. Specifically, we allow the impact of sovereign spreads on the interest–growth differential to differ across two dimensions: whether spreads are above or below a data-driven threshold k* (1.23%, estimated via minimum-RSS grid search), and whether the observation falls before or after 2010, a plausible breakpoint reflecting regime shifts in euro area macro-financial governance (Hansen, 1999). The final estimated specification includes a triple interaction between spreads, a high-spread dummy (equal to one when spreads exceed k*), and a post-2010 regime indicator, alongside controls for debt dynamics, fiscal stance and macro variables.
The estimated equation takes the form:

$$sb_t = \alpha + \beta 1 \cdot spr_t + \beta 2 \cdot 1\{spr_t > k*\} + \beta 3 \cdot (spr_t \times D2010) + \beta 4 \cdot [1\{spr_t > k*\} \times D2010] + \gamma' Z_t + \varepsilon_t$$

	Estimate	SE_OLS	t_OLS	p_OLS	SE_HAC	t_HAC	p_HAC
const	19.8	6.1006	3.2456	0.0054325	4.404	4.4959	0.00042679
spr	3.0948	0.72128	4.2908	0.00064363	0.15283	20.25	2.6348e-12
spr_high	-0.52139	0.7283	-0.71591	0.48504	0.26658	-1.9559	0.069368
spr×D2010	-2.4089	0.90126	-2.6728	0.017382	0.35982	-6.6947	7.17e-06
spr_high×D2010	0.64448	0.93159	0.6918	0.49963	0.30111	2.1403	0.049167
debtLag	-0.13303	0.048275	-2.7556	0.014719	0.033459	-3.9758	0.0012179
pb	0.88804	0.23651	3.7547	0.0019124	0.18921	4.6935	0.0002884
g	-0.39627	0.18294	-2.1661	0.046824	0.094158	-4.2085	0.00075972
pi	-0.42899	0.3383	-1.2681	0.22411	0.14917	-2.8759	0.011546

The estimation yields important and policy-relevant insights. First, the coefficient on the interaction between spreads and the post-2010 dummy is significantly negative, indicating that the sensitivity of the snowball effect to spreads diminished in the post-crisis era remaining consistent with the previous hypothesis about the importance of ECB's LLR[6] role in effectively weakening the market's ability to transmit stress into sovereign refinancing costs. Interestingly, this attenuation is not only statistically robust but also economically sizable, suggesting that euro area reforms materially altered the mechanics of debt sustainability.

When examining whether this change was coupled with nonlinear effects, meaning that we check whether spreads above a critical threshold induce larger deterioration in r–g, we find out that the high spread dummy (*spr_high*) enters negatively (–0.52) but with borderline statistical significance ($p \approx 0.069$), and somewhat counterintuitively, the magnitude of the coefficient is smaller than the baseline spread effect. This challenges the notion, advanced in some crisis literature (e.g., Corsetti et al., 2014), that sovereign stress becomes self-reinforcing above a tipping point. Instead, the point estimates imply that the pass-through from spreads to r-g may actually flatten when spreads are very high, rather than escalating. This result is potentially attributable to the anticipatory behavior of monetary authorities and the credibility of ECB backstops in high-stress states (Pietrunti & Zaghini, 2020).

Post-2010, the combined spread coefficients (including interactions) show that the r–g sensitivity stabilizes at a lower level regardless of spread intensity. For spreads below the threshold, the estimated slope drops from +3.1 pre-2010 to about +0.69 post-2010; for spreads above the threshold, the effect is similarly muted (+0.81). Hence, we find no evidence of convex amplification in spread pass-through, contrary to models of explosive debt dynamics, and instead document a pronounced flattening, both in absolute terms and in response to macro-financial regime change. The fact that the high-spread dummy is less statistically significant further reinforces the idea that once spreads breach a critical level, their additional effect on r–g may be politically or institutionally dampened, especially in countries like Italy where ECB intervention looms large.

Our findings support a narrative in which the sovereign risk channel remains relevant but is mediated by institutional evolution. While earlier literature identifies nonlinear spread effects consistent with panic-driven debt spirals, these mechanisms appear to have weakened in the post-2010 institutional regime. In line with Corsetti et al. (2014), spreads may have historically amplified r–g at high levels, but ECB backstops likely mitigated this dynamic in the later period. The transition from market discipline to policy containment mechanisms, particularly post-2010, reshaped the link between investor sentiment and fiscal sustainability, diminishing the potential for nonlinear spirals and reinforcing the stabilizing role of supranational monetary authority.

[6] Lender of Last Resort